

MH1101 Tutorial 3 (Week 4)

Solution

Reference: Sections 1.6, 2.1, 2.2, 2.3 (Lecture Notes)

1. Determine whether each integral is convergent or divergent. Evaluate those that are convergent.

(i) $\int_{-\infty}^0 \frac{1}{3-4x} dx$

(ii) $\int_e^{\infty} \frac{1}{x(\ln x)^3} dx$

(iii) $\int_{-\infty}^{\infty} (x^3 - 3x^2) dx$

(iv) $\int_{-\infty}^{\infty} xe^{-x^2} dx$

(v) $\int_{-2}^3 \frac{1}{x^4} dx$

Solution

(i) **Divergent.** Let $u = 3 - 4x$. Then $du = -4 dx \implies dx = -\frac{1}{4} du$. So by the Substitution Rule,

$$\int \frac{1}{3-4x} dx = -\frac{1}{4} \int \frac{1}{u} du = -\frac{1}{4} \ln |u| + C = -\frac{1}{4} \ln |3-4x| + C.$$

$$\begin{aligned} \int_{-\infty}^0 \frac{1}{3-4x} dx &= \lim_{t \rightarrow -\infty} \int_t^0 \frac{1}{3-4x} dx \\ &= \lim_{t \rightarrow -\infty} \left[-\frac{1}{4} \ln |3-4x| \right]_t^0 \\ &= \lim_{t \rightarrow -\infty} \left(-\frac{1}{4} \ln 3 + \frac{1}{4} \ln |3-4t| \right) = \infty. \end{aligned}$$

- (ii) **Convergent.** Let $u = \ln x$. Then $du = \frac{1}{x} dx$. Moreover, if $x = e$ then $u = \ln e = 1$, and if $x = t$ then $u = \ln t$. It follows that

$$\begin{aligned}
 \int_e^\infty \frac{1}{x(\ln x)^3} dx &= \lim_{t \rightarrow \infty} \int_e^t \frac{1}{x(\ln x)^3} dx \\
 &= \lim_{t \rightarrow \infty} \int_1^{\ln t} \frac{1}{u^3} du \\
 &= \lim_{t \rightarrow \infty} \left[\frac{u^{-2}}{-2} \right]_1^{\ln t} \\
 &= \lim_{t \rightarrow \infty} \left(-\frac{1}{2(\ln t)^2} + \frac{1}{2} \right) \\
 &= 0 + \frac{1}{2} = \frac{1}{2}.
 \end{aligned}$$

- (iii) **Divergent.** Assume for a contradiction that it is convergent. Then

$$\int_{-\infty}^\infty (x^3 - 3x^2) dx = \underbrace{\int_{-\infty}^0 (x^3 - 3x^2) dx}_{I_1} + \underbrace{\int_0^\infty (x^3 - 3x^2) dx}_{I_2},$$

where both I_1 and I_2 are convergent. But

$$\begin{aligned}
 I_1 &= \lim_{t \rightarrow -\infty} \int_t^0 (x^3 - 3x^2) dx \\
 &= \lim_{t \rightarrow -\infty} \left[\frac{x^4}{4} - x^3 \right]_t^0 \\
 &= \lim_{t \rightarrow -\infty} \left(t^3 - \frac{t^4}{4} \right) \\
 &= -\infty
 \end{aligned}$$

which means that I_1 is divergent. But this is a contradiction. So the given integral is divergent.

- (iv) **Convergent.**

$$\int_{-\infty}^\infty x e^{-x^2} dx = \int_{-\infty}^0 x e^{-x^2} dx + \int_0^\infty x e^{-x^2} dx.$$

Note that an antiderivative of xe^{-x^2} is $-\frac{1}{2}e^{-x^2}$. (You can verify this by substitution $u = -x^2$ or by directly differentiating $-\frac{1}{2}e^{-x^2}$ via Chain Rule). Thus,

$$\begin{aligned}\int_{-\infty}^0 xe^{-x^2} dx &= \lim_{t \rightarrow -\infty} \int_t^0 xe^{-x^2} dx \\ &= \lim_{t \rightarrow -\infty} \left[-\frac{1}{2}e^{-x^2} \right]_t^0 \\ &= \lim_{t \rightarrow -\infty} \left(-\frac{1}{2} \right) (1 - e^{-t^2}) \\ &= -\frac{1}{2}.\end{aligned}$$

$$\begin{aligned}\int_0^{\infty} xe^{-x^2} dx &= \lim_{t \rightarrow \infty} \int_0^t xe^{-x^2} dx \\ &= \lim_{t \rightarrow \infty} \left[-\frac{1}{2}e^{-x^2} \right]_0^t \\ &= \lim_{t \rightarrow \infty} \left(-\frac{1}{2} \right) (e^{-t^2} - 1) \\ &= \frac{1}{2}\end{aligned}$$

Hence,

$$\int_{-\infty}^{\infty} xe^{-x^2} dx = -\frac{1}{2} + \frac{1}{2} = 0.$$

(v) **Divergent.**

The function $\frac{1}{x^4}$ has a discontinuity at $x = 0$ inside the interval $[-2, 3]$. So the integral is equal to (if it exists)

$$\int_{-2}^3 \frac{1}{x^4} dx = \int_{-2}^0 \frac{1}{x^4} dx + \int_0^3 \frac{1}{x^4} dx$$

But

$$\begin{aligned}\int_{-2}^0 \frac{1}{x^4} dx &= \lim_{t \rightarrow 0^-} \left[\frac{x^{-3}}{-3} \right]_{-2}^t \\ &= \lim_{t \rightarrow 0^-} \left(-\frac{1}{3t^3} - \frac{1}{24} \right) = \infty.\end{aligned}$$

So $\int_{-2}^0 \frac{1}{x^4} dx$ is divergent. Thus, the given integral is divergent.

□

2. Suppose both $\int_{-\infty}^c f(x) dx$ and $\int_c^{\infty} f(x) dx$ are convergent for any real number c . Show that for any real number a and b ,

$$\int_{-\infty}^a f(x) dx + \int_a^{\infty} f(x) dx = \int_{-\infty}^b f(x) dx + \int_b^{\infty} f(x) dx.$$

Solution Assume without loss of generality that $a < b$.

$$\begin{aligned}& \int_{-\infty}^a f(x) dx + \int_a^{\infty} f(x) dx \\ &= \lim_{t \rightarrow -\infty} \int_t^a f(x) dx + \lim_{u \rightarrow \infty} \int_a^u f(x) dx \\ &= \lim_{t \rightarrow -\infty} \int_t^a f(x) dx + \lim_{u \rightarrow \infty} \left(\int_a^b f(x) dx + \int_b^u f(x) dx \right) \\ &= \lim_{t \rightarrow -\infty} \int_t^a f(x) dx + \int_a^b f(x) dx + \lim_{u \rightarrow \infty} \int_b^u f(x) dx \\ &= \lim_{t \rightarrow -\infty} \left(\int_t^a f(x) dx + \int_a^b f(x) dx \right) + \lim_{u \rightarrow \infty} \int_b^u f(x) dx \\ &= \lim_{t \rightarrow -\infty} \int_t^b f(x) dx + \lim_{u \rightarrow \infty} \int_b^u f(x) dx \\ &= \int_{-\infty}^b f(x) dx + \int_b^{\infty} f(x) dx\end{aligned}$$

From the second step to the third step above, note that we have made the replacement

$$\lim_{t \rightarrow \infty} \int_a^b f(x) dx = \lim_{t \rightarrow -\infty} \int_a^b f(x) dx,$$

which is okay since $\int_a^b f(x) dx$ is a constant, and so it does not depend on t , and so $\lim_{t \rightarrow \infty}$ or $\lim_{t \rightarrow -\infty}$ does not change $\int_a^b f(x) dx$. In fact,

$$\lim_{t \rightarrow \infty} \int_a^b f(x) dx = \lim_{t \rightarrow -\infty} \int_a^b f(x) dx = \int_a^b f(x) dx.$$

The result justifies our definition of $\int_{-\infty}^{\infty} f(x) dx$ in the lecture, where any real number a can be used to evaluate the integral:

$$\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^a f(x) dx + \int_a^{\infty} f(x) dx.$$

□

3. Sketch the region enclosed by the given curves, and find its area.

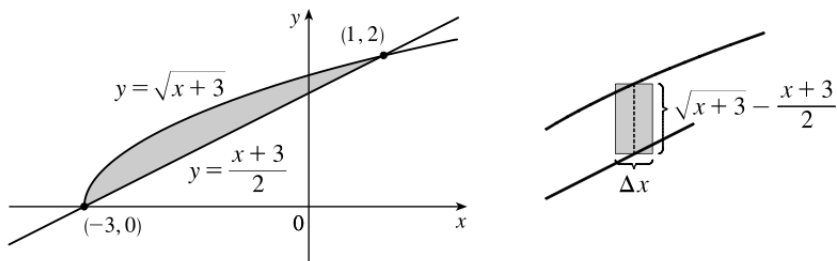
(i) $y = \sqrt{x+3}$, $y = (x+3)/2$.

(ii) $4x + y^2 = 12$, $x = y$.

(iii) $y = \cos x$, $y = \sin 2x$, $x = 0$, $x = \frac{\pi}{2}$.

Solution

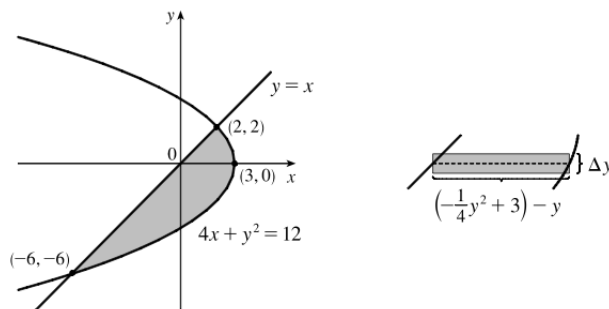
- (i) First, find the points of intersection: $\sqrt{x+3} = (x+3)/2 \implies (\sqrt{x+3})^2 = ((x+3)/2)^2 \implies 4(x+3) = (x+3)^2 \implies (x+3)((x+3) - 4) = 0 \implies (x+3)(x-1) = 0 \implies x = 1, -3$. The intersection points are $(1, 2)$ and $(-3, 0)$.



The area is

$$\begin{aligned}
 A &= \int_{-3}^1 \left(\sqrt{x+3} - \frac{x+3}{2} \right) dx \\
 &= \left[\frac{2}{3}(x+3)^{3/2} - \frac{(x+3)^2}{4} \right]_{-3}^1 \\
 &= \left(\frac{16}{3} - 4 \right) - (0 - 0) \\
 &= \frac{4}{3}.
 \end{aligned}$$

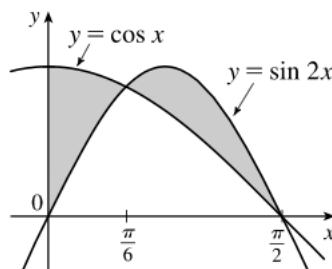
- (ii) First, find the points of intersection: $4x + x^2 = 12 \implies (x+6)(x-2) = 0 \implies x = -6, 2$. The intersection points are $(-6, -6)$, $(2, 2)$.



The area is

$$\begin{aligned}
 A &= \int_{-6}^2 \left(\left(3 - \frac{y^2}{4} \right) - y \right) dy = \left[3y - \frac{y^3}{12} - \frac{y^2}{2} \right]_{-6}^2 \\
 &= \left(6 - \frac{2}{3} - 2 \right) - (-18 + 18 - 18) = \frac{64}{3}.
 \end{aligned}$$

- (iii) First, we find the points of intersection. Notice that $\cos x = \sin 2x = 2 \sin x \cos x \iff \cos x(2 \sin x - 1) = 0 \iff \cos x = 0$ or $\sin x = \frac{1}{2}$. Since, $0 \leq x \leq \frac{\pi}{2}$, there are two solutions: $x = \frac{\pi}{6}$, $x = \frac{\pi}{2}$.



The area of the shaded region is equal to

$$\begin{aligned}
 A &= \int_0^{\pi/6} (\cos x - \sin 2x) dx + \int_{\pi/6}^{\pi/2} (\sin 2x - \cos x) dx \\
 &= \left[\sin x + \frac{\cos 2x}{2} \right]_0^{\pi/6} + \left[-\frac{\cos 2x}{2} - \sin x \right]_{\pi/6}^{\pi/2} \\
 &= \left(\frac{1}{2} + \frac{1}{4} \right) - \left(0 + \frac{1}{2} \right) + \left(\frac{1}{2} - 1 \right) - \left(-\frac{1}{4} - \frac{1}{2} \right) \\
 &= \frac{1}{2}.
 \end{aligned}$$

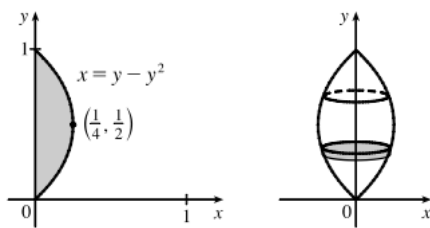
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4. Using the disk-washer method, find the volume of the solid obtained by rotating the region bounded by the curves about the specified line.

- (i) $x = y - y^2$, $x = 0$; about the y -axis.
- (ii) $y = \frac{1}{4}x^2$, $y = 5 - x^2$; about the x -axis.
- (iii) $y = x^2$, $y^2 = 8x$; about the line $x = 2$.

Solution

- (i) A (horizontal) cross-section is a disk with radius $y - y^2$, so its area is $A(y) = \pi(y - y^2)^2$:

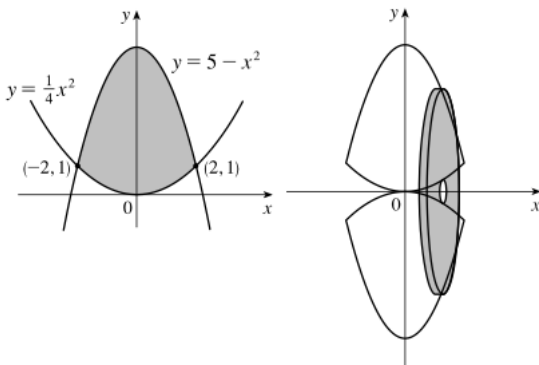


By the Disk-Washer method, the required volume is

$$\begin{aligned}
 V &= \int_0^1 A(y) \, dy = \int_0^1 \pi(y - y^2)^2 \, dy \\
 &= \pi \int_0^1 y^2 - 2y^3 + y^4 \, dy \\
 &= \pi \left[\frac{y^3}{3} - \frac{y^4}{2} + \frac{y^5}{5} \right]_0^1 = \frac{\pi}{30}.
 \end{aligned}$$

(ii) A (vertical) cross-section is a washer whose area is

$$A(x) = \pi(5 - x^2)^2 - \pi \left(\frac{x^2}{4} \right)^2 = 25 - 10x^2 + \frac{15}{16}x^4.$$

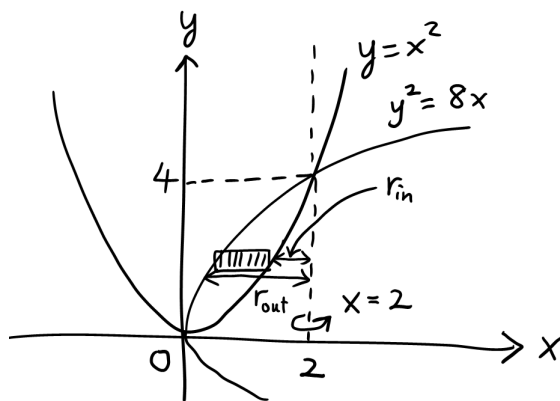


By the Disk-Washer method,

$$\begin{aligned}
 V &= \int_{-2}^2 A(x) dx = \int_{-2}^2 \pi \left(25 - 10x^2 + \frac{15}{16}x^4 \right) dx \\
 &= \pi \left[25x - \frac{10x^3}{3} + \frac{3x^5}{16} \right]_{-2}^2 \\
 &= \pi \left(\left(50 - \frac{80}{3} + 6 \right) - \left(-50 + \frac{80}{3} - 6 \right) \right) \\
 &= \frac{176\pi}{3}.
 \end{aligned}$$

(iii) Slide the solid perpendicular to the axis of rotation ($x = 2$). The cross-sections are horizontal washers, for $0 \leq y \leq 4$, with

- outer radius $r_{out} = 2 - \frac{y^2}{8}$
- inner radius $r_{in} = 2 - \sqrt{y}$

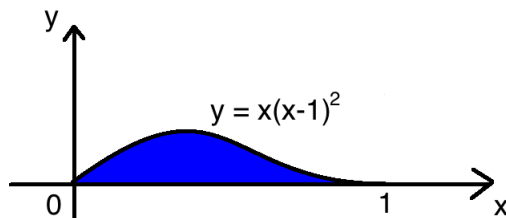


Hence, the volume is

$$\begin{aligned}
 V &= \int_0^4 A(y) dy \\
 &= \int_0^4 \pi(r_{out}^2 - r_{in}^2) dy \\
 &= \pi \int_0^4 \left(2 - \frac{y^2}{8}\right)^2 - (2 - \sqrt{y})^2 dy \\
 &= \pi \int_0^4 \left(4 - \frac{y^2}{2} + \frac{y^4}{64}\right) - (4 - 4\sqrt{y} + y) dy \\
 &= \pi \int_0^4 \left(-\frac{y^2}{2} + \frac{y^4}{64} + 4y^{1/2} - y\right) dy \\
 &= \pi \left[-\frac{y^3}{6} + \frac{y^5}{320} + \frac{4y^{3/2}}{3/2} - \frac{y^2}{2}\right]_0^4 \\
 &= \pi \left(-\frac{64}{6} + \frac{1024}{320} + \frac{8}{3}(8) - \frac{16}{2}\right) \\
 &= \frac{88\pi}{15}.
 \end{aligned}$$

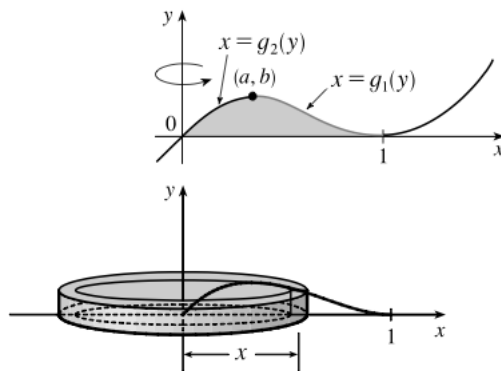
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5. Let S be the solid obtained by rotating the region shown in the figure about the y -axis. Explain why it is awkward to use the disk-washer method to find the volume V of S . Sketch a typical approximating shell. What are its circumference and height? Use cylindrical shells to find V .



Solution If we were to use the Disk-Washer method, we would first have

to locate the local maximum point (a, b) of $y = x(x - 1)^2$. Then we would need to split the curve on the left-hand side and right-hand side of (a, b) and express them as functions of y : $x = g_1(y)$, $x = g_2(y)$ (see figure below). This step would be difficult as it involves the cubic formula. Finally, we also need to find the volume using $V = \int_0^b \pi((g_1(y))^2 - (g_2(y))^2) dy$. This is just brutal!



On the other hand, using shells, a typical cylindrical shell has radius x , so its circumference is $2\pi x$. Its height is y , which is $x(x - 1)^2$. So the volume is

$$\begin{aligned} V &= \int_0^1 2\pi x(x(x - 1)^2) dx = 2\pi \int_0^1 x^2(x^2 - 2x + 1) dx \\ &= 2\pi \left[\frac{x^5}{5} - \frac{x^4}{2} + \frac{x^3}{3} \right]_0^1 = \frac{\pi}{15}. \end{aligned}$$

This is just a walk in the park! □

Answer Keys. 1. (i) Divergent (ii) Convergent (iii) Divergent (iv) Convergent (v) Divergent 3. (i) $\frac{4}{3}$ (ii) $\frac{64}{3}$ (iii) $\frac{1}{2}$ 4. (i) $\frac{\pi}{30}$ (ii) $\frac{176\pi}{3}$ (iii) $\frac{88\pi}{15}$ 5. $\frac{\pi}{15}$